Data Preprocessing

Import

In [36]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import ttest_rel, f_oneway, tukey_hsd, pearsonr

reviews = pd.read_csv("rotten_tomatoes_movie_reviews.csv")
movies = pd.read_csv("rotten_tomatoes_movies.csv")
```

Parse

In [37]:
```
# We will start parsing by removing duplicate movies and reviews
movies = movies.drop_duplicates(subset='id')

# We turn Box Office sales from strings with letters (e.g 'M' for million) into dol
movies['boxOffice'] = movies['boxOffice'].str.replace('$', '').str.replace('M', 'e6
movies['boxOffice'] = pd.to_numeric(movies['boxOffice'], errors='coerce')

# Now we convert release dates and review dates from text to datetime
movies['releaseDateTheaters'] = pd.to_datetime(movies['releaseDateTheaters'], error
movies['releaseDateStreaming'] = pd.to_datetime(movies['releaseDateStreaming'], err

# The genre can be a list of multiple genres, so we take the first one as the movie
movies['primaryGenre'] = movies['genre'].str.split(',').str[0]

# Define the difference in scores
movies['scoreGap'] = movies['audienceScore'] - movies['tomatoMeter']

movies[['title', 'tomatoMeter', 'audienceScore', 'scoreGap', 'primaryGenre', 'boxOf
```

Out[37]:

	title	tomatoMeter	audienceScore	scoreGap	primaryGenre	boxOffice
0	Space Zombie Bingo!	NaN	50.0	NaN	Comedy	NaN
1	The Green Grass	NaN	NaN	NaN	Drama	NaN
2	Love, Lies	NaN	43.0	NaN	Drama	NaN
3	Sore Losers	NaN	60.0	NaN	Action	NaN
4	Dinosaur Island	NaN	70.0	NaN	Fantasy	NaN

In [38]:
```
# Removing duplicates
reviews = reviews.drop_duplicates(subset='reviewId')
reviews['creationDate'] = pd.to_datetime(reviews['creationDate'], errors='coerce')

# In the raw data, the critics' original scores are in different formats such as ("
# We can convert the fraction ones to a [0-1] scale so they can be compared (other
```

```
parts = reviews['originalScore'].str.split('/', expand=True)
reviews['scoreNorm'] = pd.to_numeric(parts[0], errors='coerce') / pd.to_numeric(par
reviews.loc[reviews['scoreNorm'] > 1, 'scoreNorm'] = np.nan

reviews['isFresh'] = reviews['reviewState'] == 'fresh'

reviews[['originalScore', 'scoreNorm', 'reviewState', 'isFresh']].head()
```

Out[38]:

	originalScore	scoreNorm	reviewState	isFresh
0	3.5/4	0.875	fresh	True
1	1/5	0.200	rotten	False
2	NaN	NaN	fresh	True
3	2.5/5	0.500	rotten	False
4	NaN	NaN	fresh	True

Organize

In [39]:

```
# Count each movie's reviews and its percentage of Fresh reviews
review_counts = reviews.groupby('id').agg(numReviews=('reviewId', 'count'), freshRa
movies = movies.merge(review_counts, on='id', how='left')
display(movies.head())

# Keep movies with both an audience and Tomatometer (critic) score
scored = movies.dropna(subset=['tomatoMeter', 'audienceScore'])
scored[['tomatoMeter', 'audienceScore', 'scoreGap', 'runtimeMinutes', 'numReviews']
```

	id	title	audienceScore	tomatoMeter	rating	ratingContents	releas
0	space-zombie-bingo	Space Zombie Bingo!	50.0	NaN	NaN	NaN	
1	the_green_grass	The Green Grass	NaN	NaN	NaN	NaN	
2	love_lies	Love, Lies	43.0	NaN	NaN	NaN	
3	the_sore_losers_1997	Sore Losers	60.0	NaN	NaN	NaN	
4	dinosaur_island_2002	Dinosaur Island	70.0	NaN	NaN	NaN	

Out[39]:

	tomatoMeter	audienceScore	scoreGap	runtimeMinutes	numReviews
count	30052.000000	30052.000000	30052.000000	29367.000000	29854.000000
mean	64.525090	62.310229	-2.214861	101.506453	44.397836
std	28.045784	20.862795	22.745236	21.090280	61.756210
min	0.000000	0.000000	-100.000000	5.000000	1.000000
25%	43.000000	47.000000	-17.000000	90.000000	9.000000
50%	71.000000	65.000000	-4.000000	98.000000	19.000000
75%	88.000000	79.000000	11.000000	110.000000	51.000000
max	100.000000	100.000000	100.000000	584.000000	597.000000

Data Exploration

Hypothesis 1: Is there a difference between audience and critic scores?

> H0: There is no difference between audience and critic scores. The mean of (audience score - critic score) is 0.

> HA: There is a difference between audience and critic scores. The mean of (audience score - critic score) is not 0.

Statistical Method: **paired t-test**

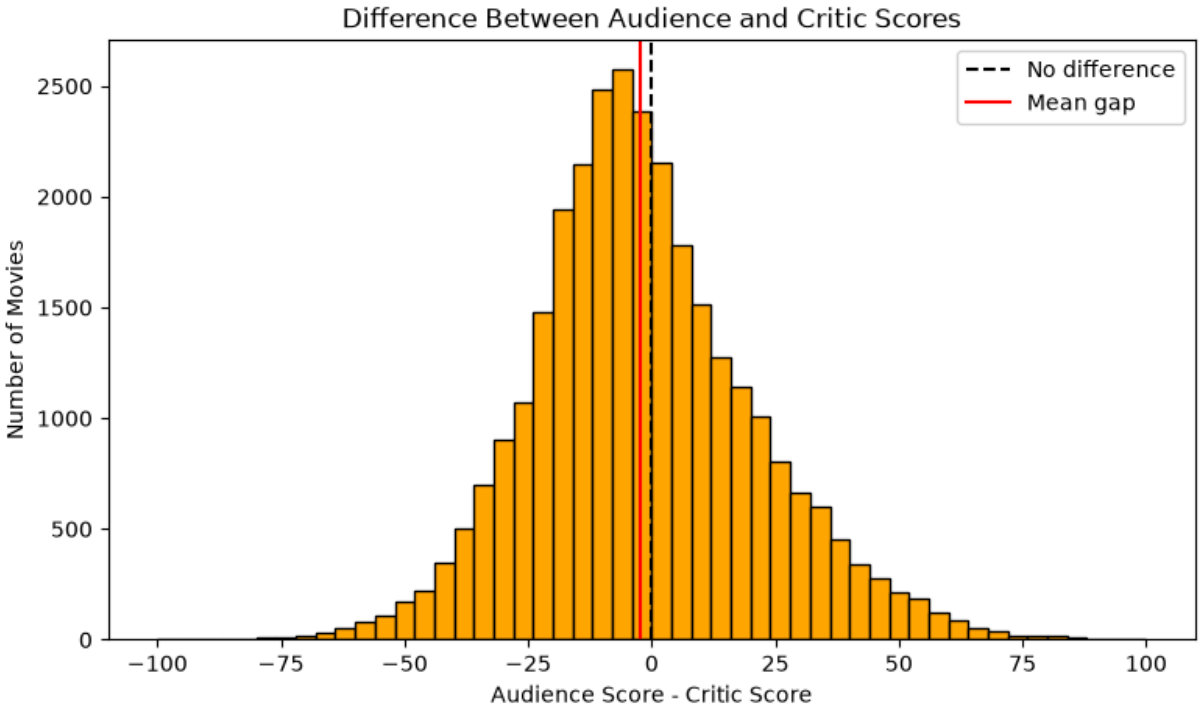
Every movie has both an audience score and a critic score, so the two samples are paired and not independent. Our sample is very large, so by the CLT the mean difference is approximately normal.

In [40]:
```
t_stat, p = ttest_rel(scored['audienceScore'], scored['tomatoMeter'])
print("mean critic score:", scored['tomatoMeter'].mean())
print("mean audience score:", scored['audienceScore'].mean())
print("mean gap:", scored['scoreGap'].mean())
print("t-statistic:", t_stat)
print("p-value:", p)
```

```
mean critic score: 64.52508984426993
mean audience score: 62.31022893651005
mean gap: -2.214860907759883
t-statistic: -16.880787311225728
p-value: 1.2249939881945988e-63
```

In [41]:
```
plt.figure(figsize=(9, 5))
plt.hist(scored['scoreGap'], bins=50, color='orange', edgecolor='black')
plt.axvline(0, color='black', linestyle='--', label='No difference')
plt.axvline(scored['scoreGap'].mean(), color='red', label='Mean gap')
```

```
plt.xlabel('Audience Score - Critic Score')
plt.ylabel('Number of Movies')
plt.title('Difference Between Audience and Critic Scores')
plt.legend()
plt.show()
```



Conclusion: The p-value is 1.22e-63, which is less than 0.05, so we reject the null hypothesis. There is a significant difference between audience and critic scores. On average, critics score movies 2.2 points higher than audiences (64.5 vs. 62.3).

Hypothesis 2: Is the gap wider in certain genres?

> H0: Genre has no effect on the gap. The mean score gap is the same for all genres.

> HA: Genre does have an effect on the gap. At least one genre has a different mean score gap.

Statistical Method: **one-way ANOVA**

We are comparing the mean of one numeric variable (score gap) across more than 2 groups. We use each movie's first genre so no movie is counted twice, and only the 6 most common genres so each group has enough movies.

In [42]:
```
top_genres = scored['primaryGenre'].value_counts().head(6).index
groups = [scored[scored['primaryGenre'] == g]['scoreGap'] for g in top_genres]

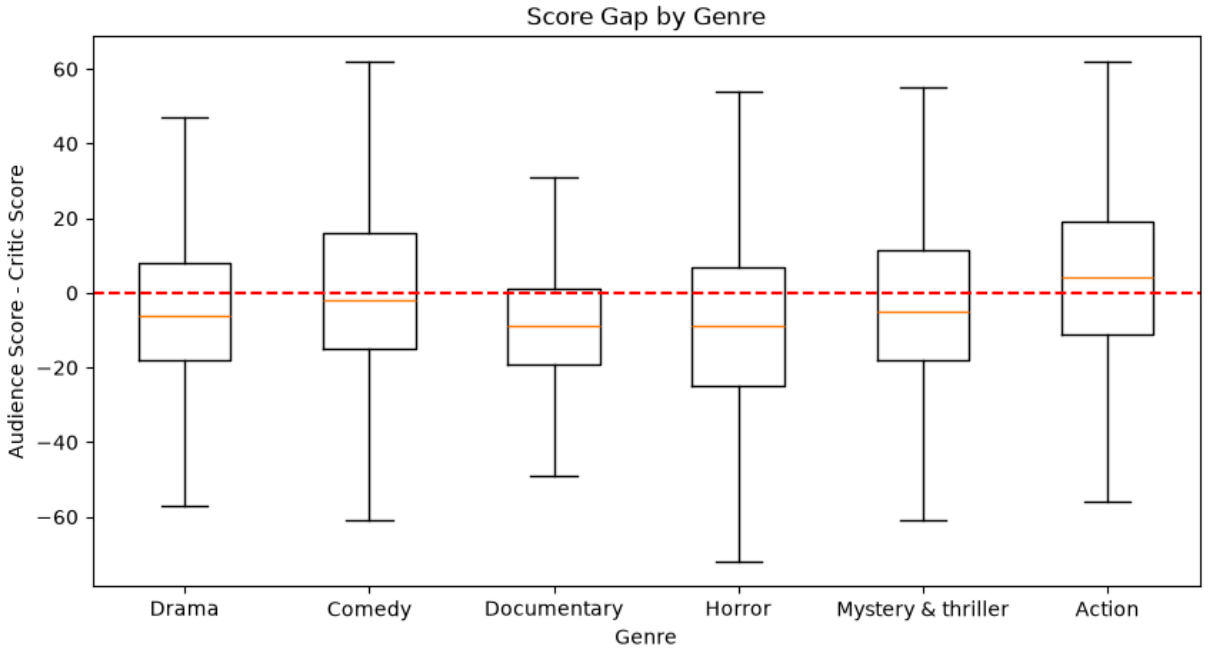
f_stat, p = f_oneway(*groups)
print(scored[scored['primaryGenre'].isin(top_genres)].groupby('primaryGenre')['scor
```

```
print("\nF-statistic:", f_stat)
print("p-value:", p)
```

```
primaryGenre
Action                 4.309567
Comedy                 0.510465
Documentary           -8.336466
Drama                 -3.541595
Horror                -8.366013
Mystery & thriller    -2.349590
Name: scoreGap, dtype: float64

F-statistic: 130.11935974850454
p-value: 2.46827876139617e-136
```

In [43]:
```
plt.figure(figsize=(10, 5))
plt.boxplot(groups, tick_labels=top_genres, showfliers=False)
plt.axhline(0, color='red', linestyle='--')
plt.xlabel('Genre')
plt.ylabel('Audience Score - Critic Score')
plt.title('Score Gap by Genre')
plt.show()
```



Since ANOVA only tells us that at least one genre is different, we use Tukey's HSD to see which pairs of genres are different (0 to 5 follow the order of `top_genres`).

In [44]:
```
print(list(top_genres))
print(tukey_hsd(*groups))
```

```
['Drama', 'Comedy', 'Documentary', 'Horror', 'Mystery & thriller', 'Action']
Pairwise Group Comparisons (95.0% Confidence Interval)
Comparison  Statistic  p-value  Lower CI  Upper CI
 (0 - 1)     -4.052     0.000    -5.209    -2.895
 (0 - 2)      4.795     0.000     3.511     6.079
 (0 - 3)      4.824     0.000     3.246     6.402
 (0 - 4)     -1.192     0.274    -2.791     0.407
 (0 - 5)     -7.851     0.000    -9.604    -6.098
 (1 - 0)      4.052     0.000     2.895     5.209
 (1 - 2)      8.847     0.000     7.496    10.197
 (1 - 3)      8.876     0.000     7.244    10.509
 (1 - 4)      2.860     0.000     1.208     4.512
 (1 - 5)     -3.799     0.000    -5.601    -1.997
 (2 - 0)     -4.795     0.000    -6.079    -3.511
 (2 - 1)     -8.847     0.000   -10.197    -7.496
 (2 - 3)      0.030     1.000    -1.695     1.754
 (2 - 4)     -5.987     0.000    -7.731    -4.243
 (2 - 5)    -12.646     0.000   -14.532   -10.760
 (3 - 0)     -4.824     0.000    -6.402    -3.246
 (3 - 1)     -8.876     0.000   -10.509    -7.244
 (3 - 2)     -0.030     1.000    -1.754     1.695
 (3 - 4)     -6.016     0.000    -7.987    -4.046
 (3 - 5)    -12.676     0.000   -14.773   -10.578
 (4 - 0)      1.192     0.274    -0.407     2.791
 (4 - 1)     -2.860     0.000    -4.512    -1.208
 (4 - 2)      5.987     0.000     4.243     7.731
 (4 - 3)      6.016     0.000     4.046     7.987
 (4 - 5)     -6.659     0.000    -8.772    -4.546
 (5 - 0)      7.851     0.000     6.098     9.604
 (5 - 1)      3.799     0.000     1.997     5.601
 (5 - 2)     12.646     0.000    10.760    14.532
 (5 - 3)     12.676     0.000    10.578    14.773
 (5 - 4)      6.659     0.000     4.546     8.772
```

Conclusion: The ANOVA p-value is 2.47e-136, which is less than 0.05, so we reject the null hypothesis. The gap between audience and critic scores depends on genre. Tukey's HSD shows 13 of 15 genre pairs are significantly different. Action has the highest mean gap, meaning audiences rate it most favorably compared to critics, followed by Comedy. Documentary and Horror have the lowest mean gaps, meaning critics rate them most favorably compared to audiences. The only pairs that are not significantly different are Drama vs. Mystery & thriller and Documentary vs. Horror.

Hypothesis 3: Is a movie's critic score related to its audience score?

> H0: There is no correlation between critic score and audience score.

> HA: There is a correlation between critic score and audience score.

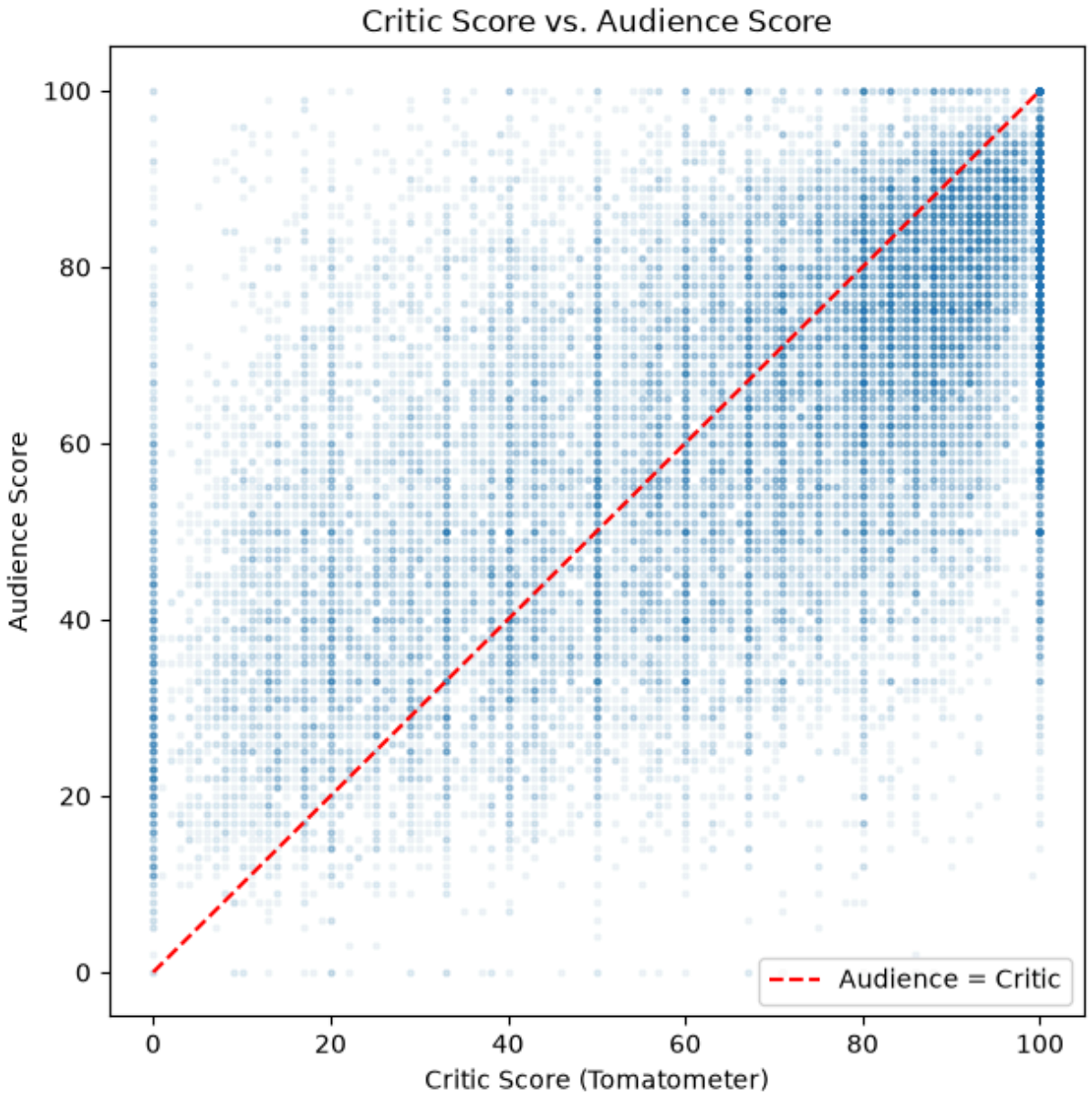
Statistical Method: **Pearson correlation**

Both variables are numeric and we want to see how strongly they're related. This matters for our plan to predict audience score from critic score later.

In [45]:
```
corr, p = pearsonr(scored['tomatoMeter'], scored['audienceScore'])
print("correlation:", corr)
print("p-value:", p)
```

```
correlation: 0.601999872548948
p-value: 0.0
```

In [46]:
```
plt.figure(figsize=(7, 7))
plt.scatter(scored['tomatoMeter'], scored['audienceScore'], alpha=0.05, s=5)
plt.plot([0, 100], [0, 100], color='red', linestyle='--', label='Audience = Critic'
plt.xlabel('Critic Score (Tomatometer)')
plt.ylabel('Audience Score')
plt.title('Critic Score vs. Audience Score')
plt.legend()
plt.show()
```



Conclusion: The p-value is so small it shows as 0.0, which is less than 0.05, so we reject the null hypothesis. There is a strong positive correlation between critic and audience scores (r = 0.60). The plot shows audiences are less extreme than critics: they score movies higher than critics when critics score low, and lower when critics score high.